

Introduction to Single-cell Transcriptomics

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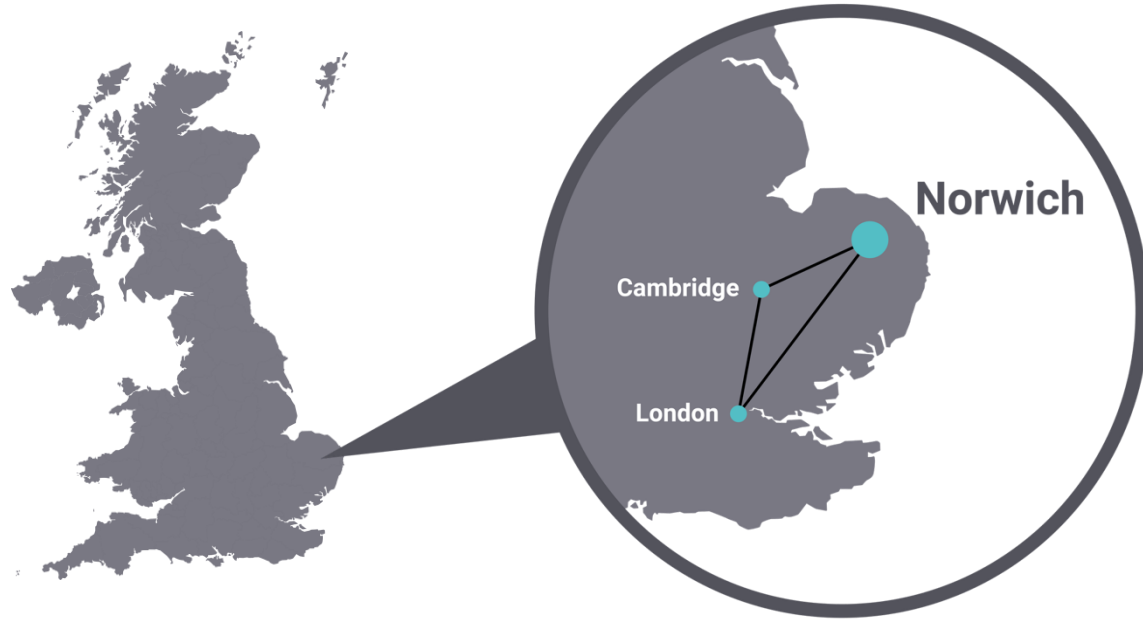
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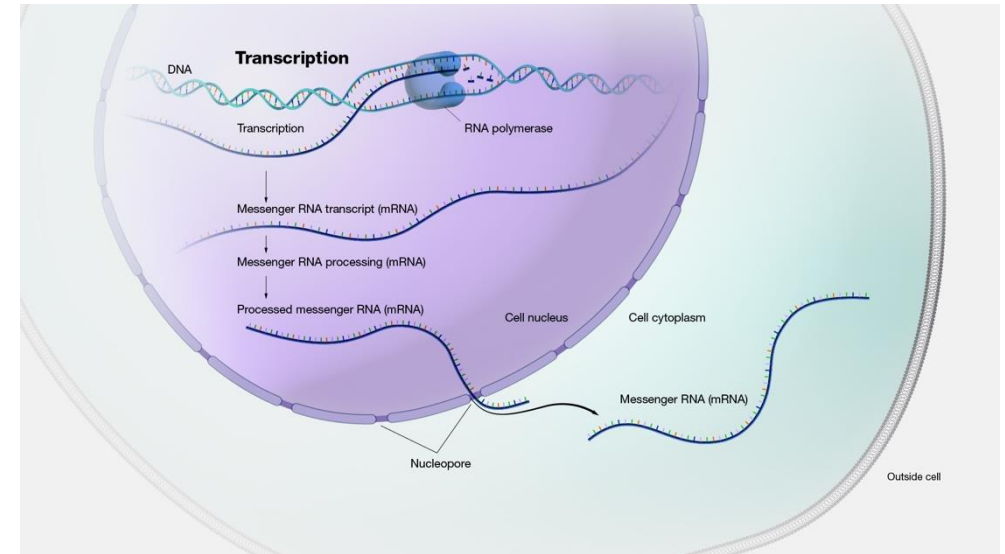
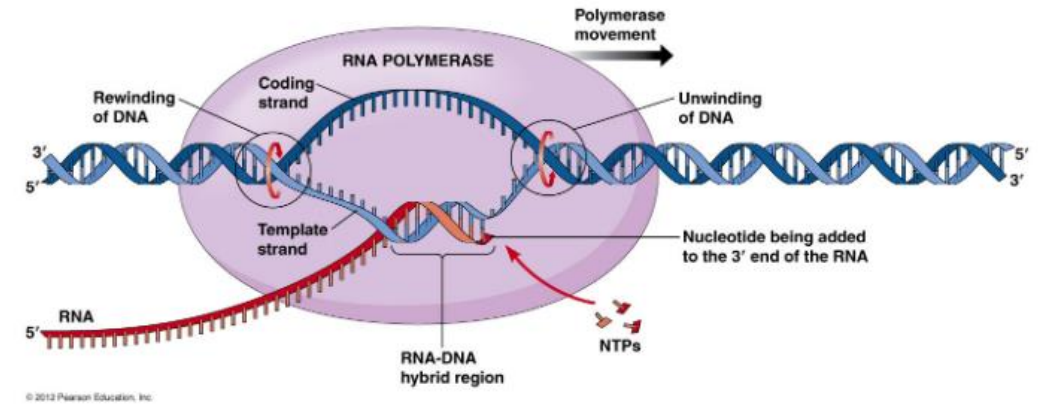
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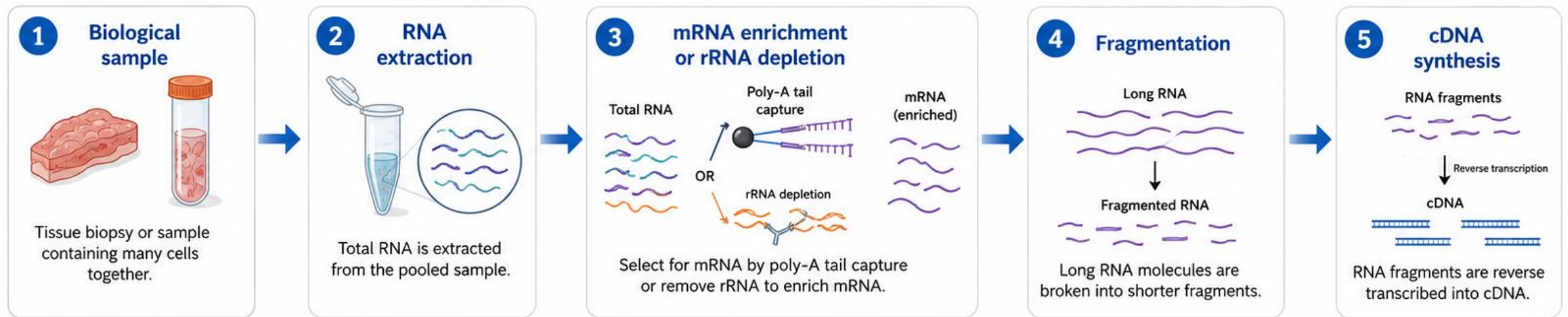
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What is transcriptomics?

- A gene is expressed when **RNA polymerase binds to its promoter** sequence and **reads the template (antisense) strand** from 3' to 5', **producing an RNA transcript** until a terminator is reached.
- The **mature mRNA molecule** migrates from the **nucleus** into the cytoplasm until it binds to a ribosome for translation into a protein.
- As **every time a gene is expressed, it produces an RNA molecule**, we can extract **the total transcribed RNA in a cell to approximate global gene expression** – this is the transcriptome.
- Transcriptomics can be used to determine **which genes are active in a cell** or tissue, and how **gene expression changes** between samples.

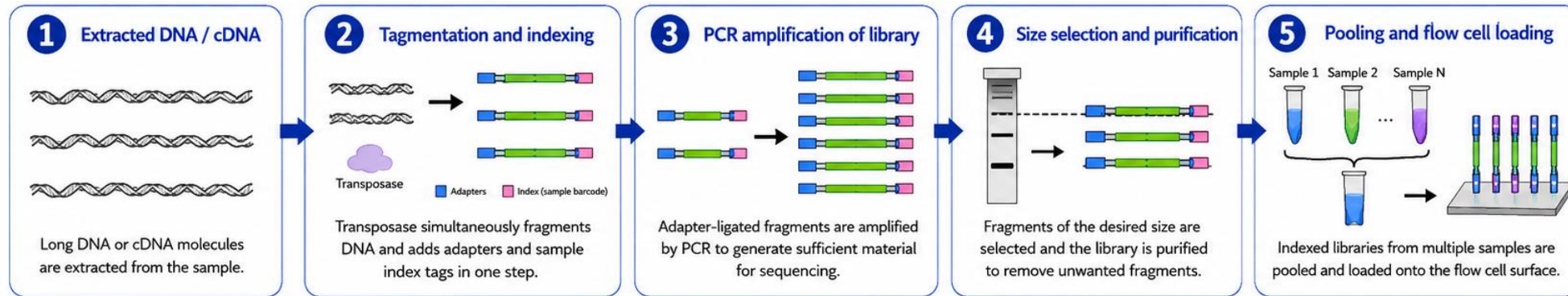


Bulk RNA-seq

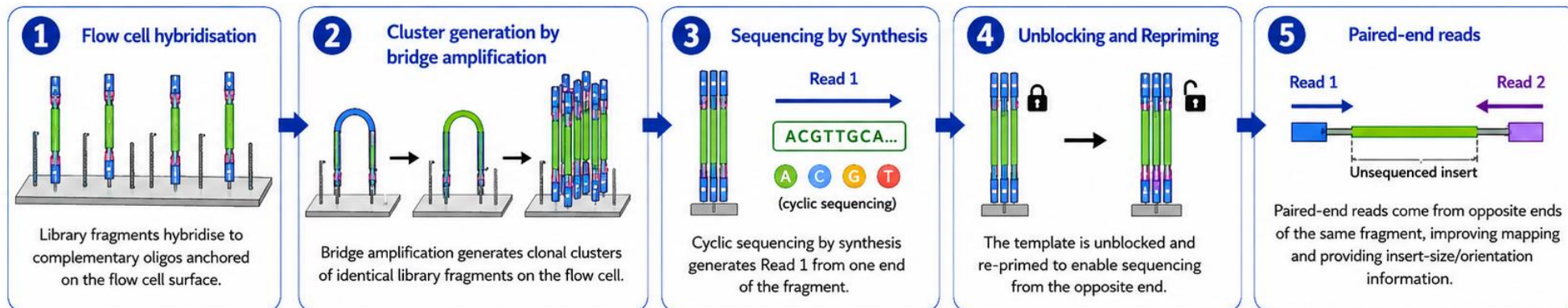


Short read sequencing

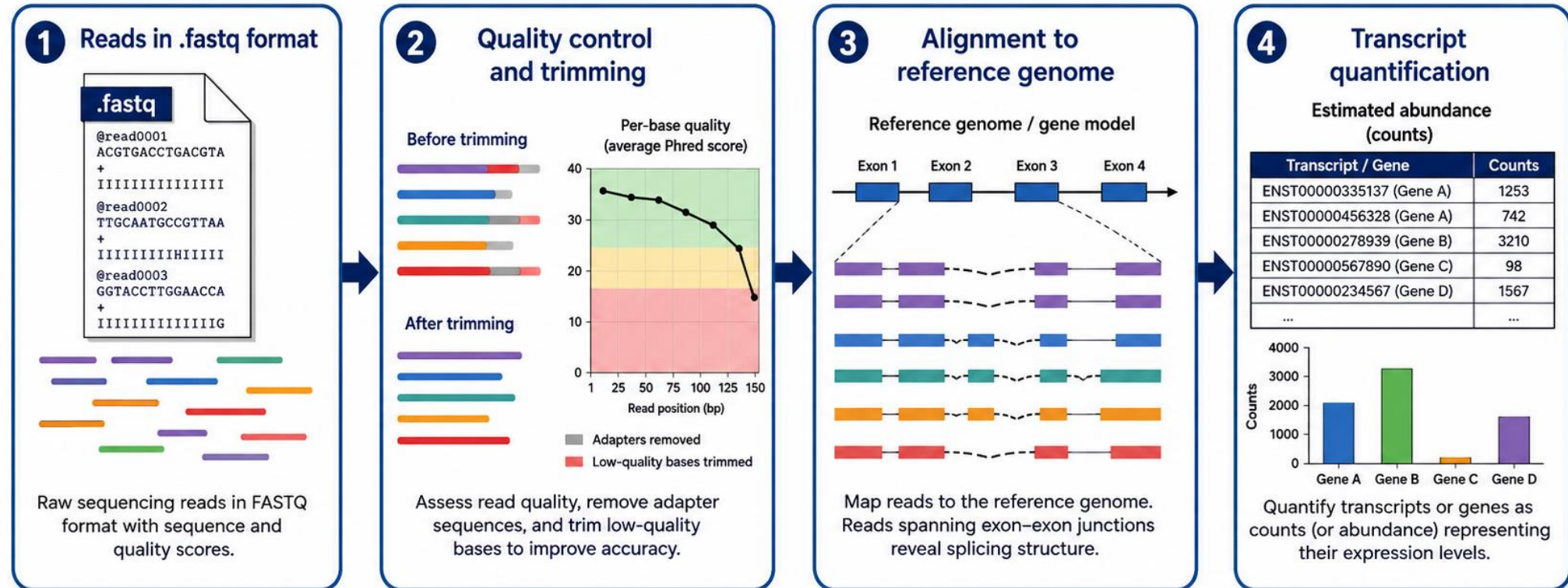
Illumina Library Prep



Paired-end Sequencing



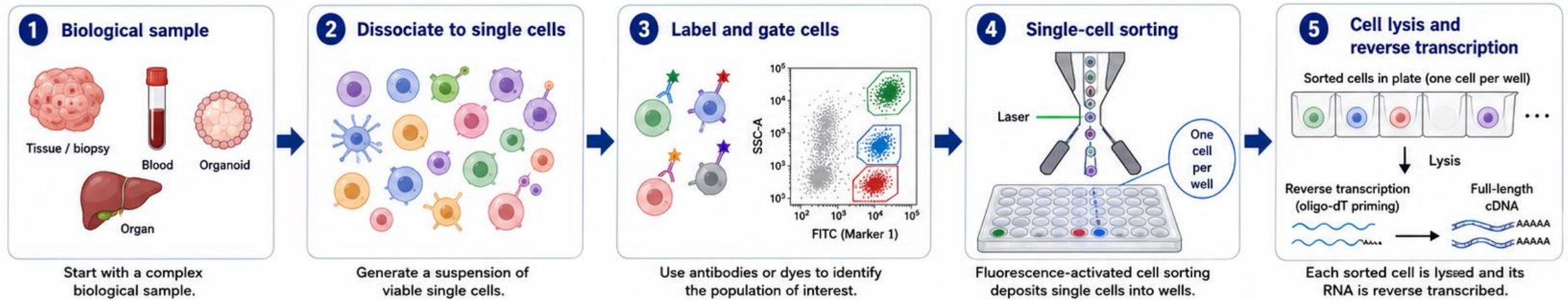
Quantifying reads



Bulk vs single-cell sequencing technology

- **Bulk sequencing pools mRNA** across an entire sample of upwards of thousands of cells
 - This has an **averaging effect** on the gene expression signal, with no resolution of cell type
 - The contributions of **rare cell types are diluted out**
 - When comparing gene expression between **non-isogenic samples (like tissue)**, it is impossible to tell whether changes in quantified transcripts are caused by **altered gene expression or proportions of cell types**
- **Single-cell sequencing resolves this** by allowing us to **partition cell before quantification** of gene expression
 - Gene expression of **individual cell types can be understood**
 - **Cell type proportions can be quantified and adjusted for between conditions** to get a better understand of mechanistically-driven changes in gene expression
- Single-cell transcriptomics allows us to ask questions that are not possible with bulk methods
 - **Which genes define each cell populations, and which cell states define a condition?**
 - **How does gene expression change** during development, disease or treatment?
 - **Cellular atlases of organs and organisms** are resources for research.

Cell sorting-based single-cell transcriptomics (Smart-seq, CEL-seq)



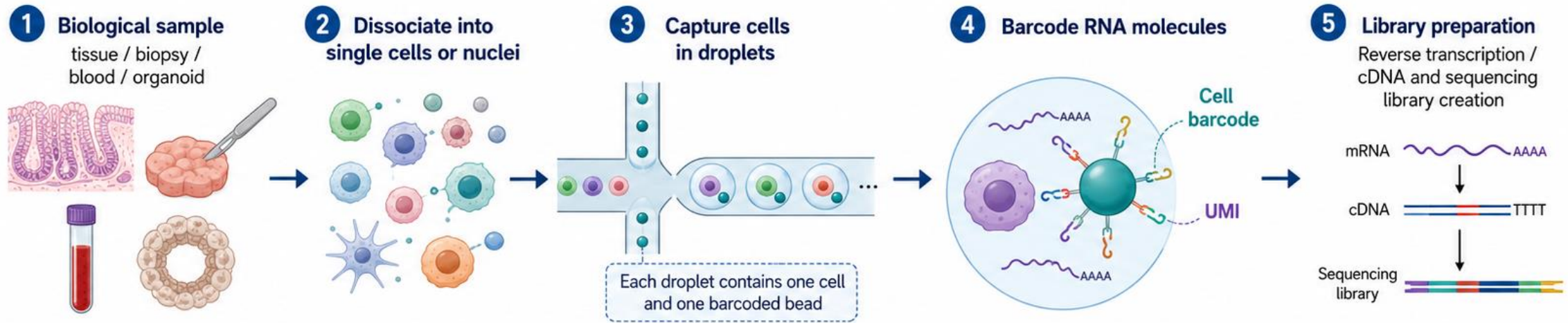
Pros

- Highly pure clusters of cells
- Full transcript information available
- More complete transcriptomes

Cons

- Low throughput (100s – 1000s of cells)
- Slow, laborious and expensive
- Must know populations of interest beforehand
- High technical variation

Droplet-based single-cell transcriptomics (10X Chromium, Drop-seq)



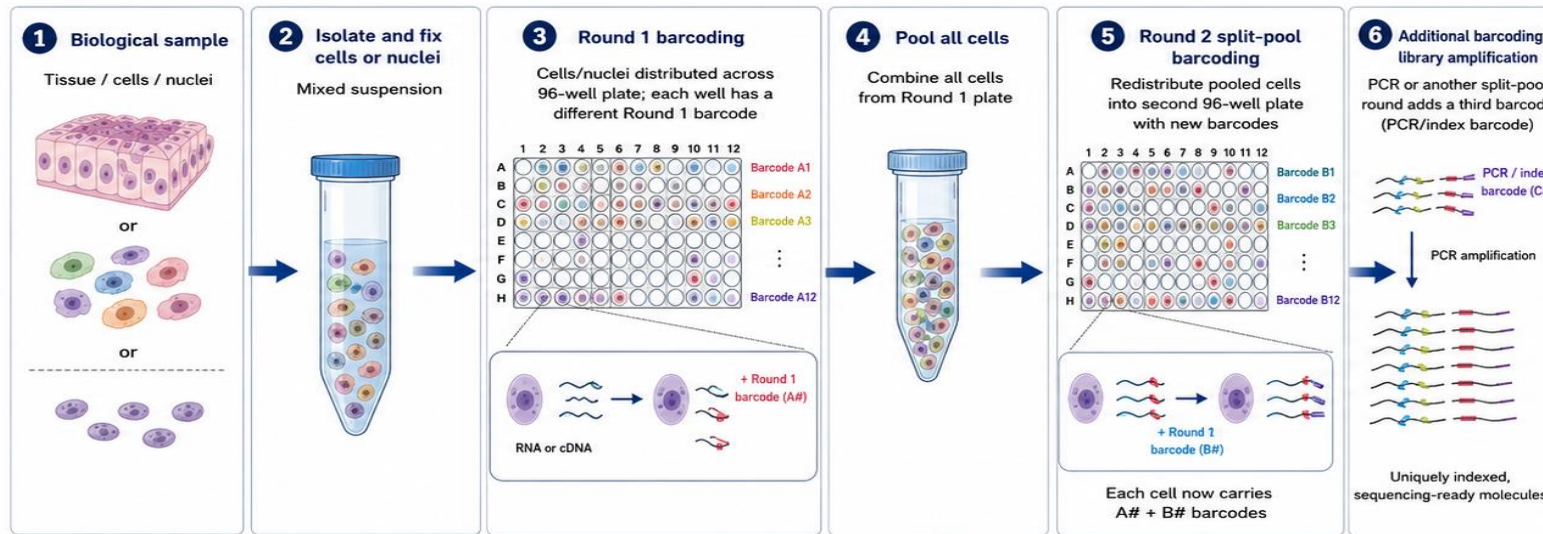
Pros

- Highly scalable (up to millions of cells)
- Cell type resolution only limited by sequencing depth
- Cheaper (per cell)
- More experimentally automated so higher throughput and reproducibility

Cons

- Capture only a small portion of the transcriptome of each cell (sparsity)
- Only 5' or 3' end data available
- Higher rates of low quality cells
- More computationally intensive

Combinatorial (split-pool) barcoding for single-cell transcriptomics (Parse Evercode)



Pros

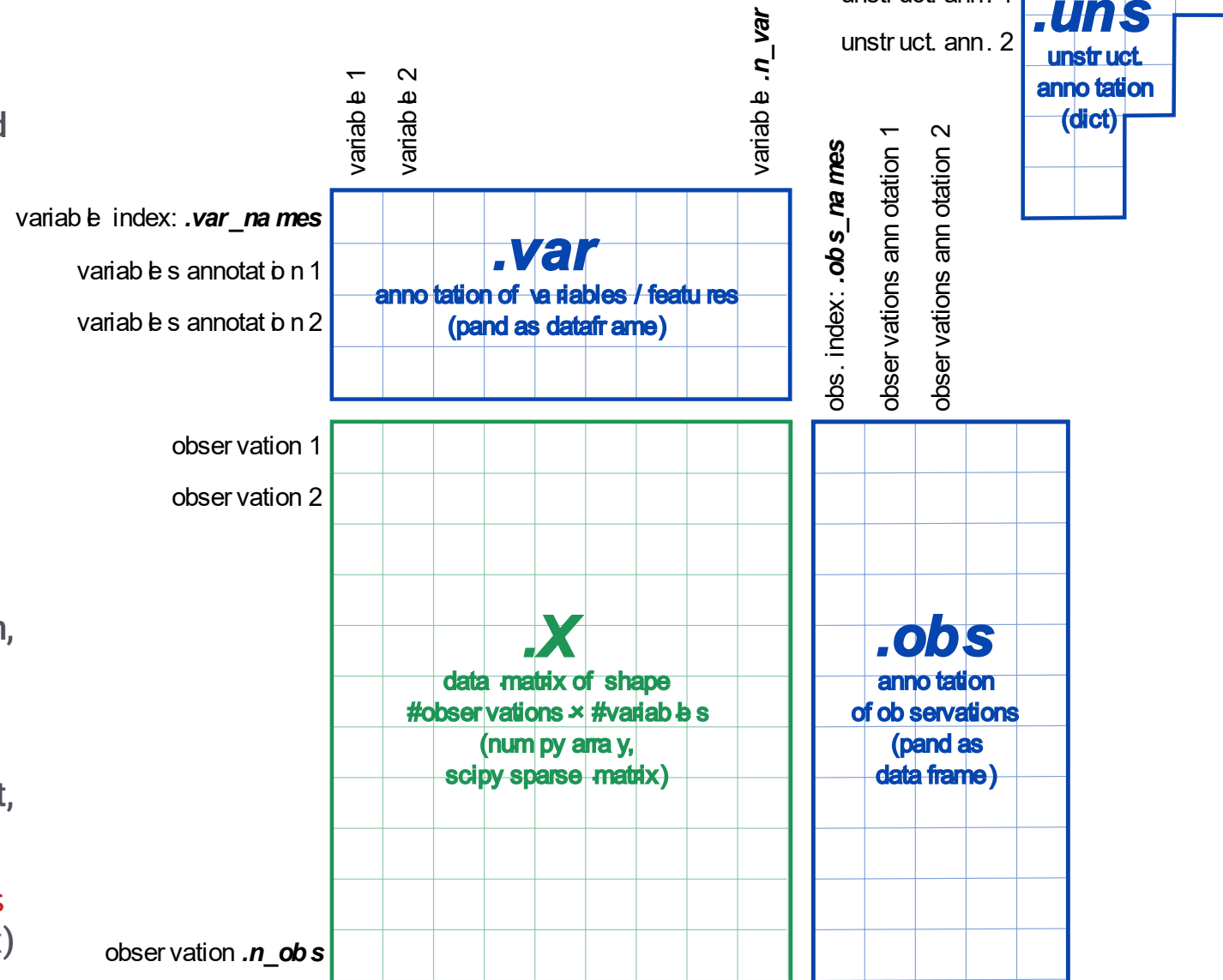
- As scalable as droplet-based methods
- No special instruments needed as it all occurs during library prep
- Compatible with paraffin-fixed cells

Cons

- Often lower sensitivity than droplet methods
- Repeated distribution, pooling and redistribution makes it experimentally complex
- Doublets via barcode collisions occur at a higher rate at high loadings

Count matrix data structure

- **AnnData** (.h5ad) is the standard Python object used to store single-cell expression data.
 - It is a structured, extensible container centred on a cell-by-gene count matrix
- Rows: cells (observations), Columns: genes (variables)
- **.X** matrix: per cell gene expression
 - Mostly zero due to sparse sampling of genes
- **.var** dataframe: gene metadata
 - Alternative genes identifiers, genomic location, gene-wise statistics, etc
- **.obs** dataframe: cell metadata
 - Parent sample, batch data, cluster assignment, cell-wise statistics, etc
- Others: **.layers** (transformed .X matrices), **.uns** (misc unstructured data), **.raw** (original .X matrix)



The single cell analysis ecosystem

Python's scVerse

- High scalability
- ML library interoperability
- Data integration and multimodal support



anndata

Standard for annotated matrices



mudata

Multimodal data format



spatialdata

Spatial data format



scanpy

Single-cell analysis framework



muon

Multi-omics analysis framework



squidpy

Spatial single-cell analysis



scvi-tools

Single-cell machine learning framework



scirpy

Single-cell immune sequencing analysis framework

R's Seurat

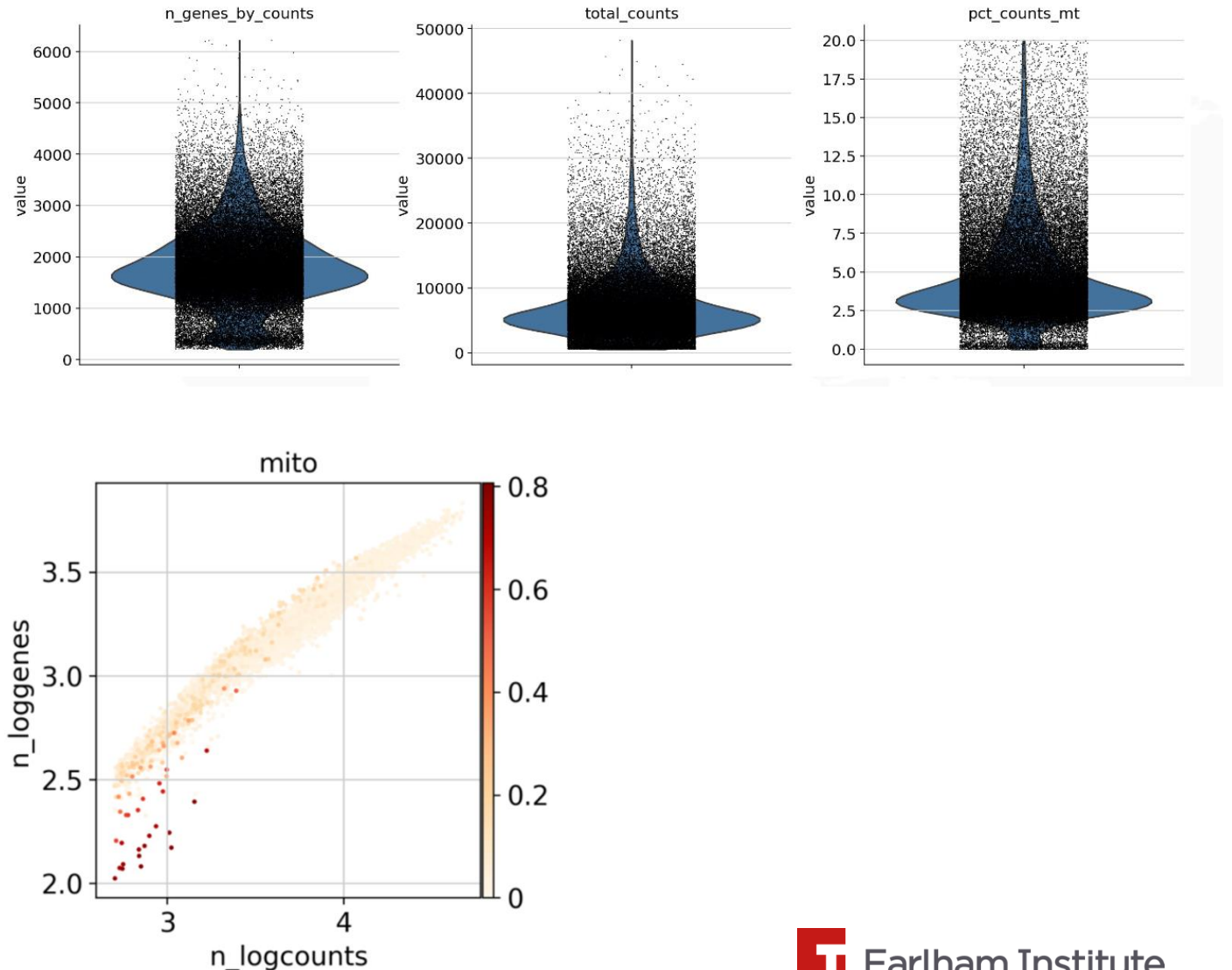
- Mature preprocessing and analysis workflows
- Easy data visualisation
- Bioconductor interoperability

Quality control for single cell data

- QC is performed on **a per-sample** basis.
- It is needed to **remove cells with that are likely to reflect technical artefacts** which would influence conclusions about biological cell types, states or disease effects.
- Low-quality cells can arise from **harsh tissue dissociation, inefficient RNA capture, ambient RNA contamination or during sample handling**
- Metrics to identify low-quality cells include:
 - **Low total transcript counts per cell** may be empty droplets, damaged cells or poorly captured cells. (>2000 UMI counts)
 - **Low number of detected genes per cell** suggest poor-quality cells with little usable RNA (>200-500 genes)
 - **Highly expressed mitochondrial or ribosomal genes** may reflect technical stress, dissociation effects, or poor sample quality (<20% MT-genes)
 - **Doublet score** to detect when two cells are captured in a single droplet (filter top X% of cells depending on expected value)
- Cells that survive filtering are trusted enough to interpret biologically

Quality control for single cell data

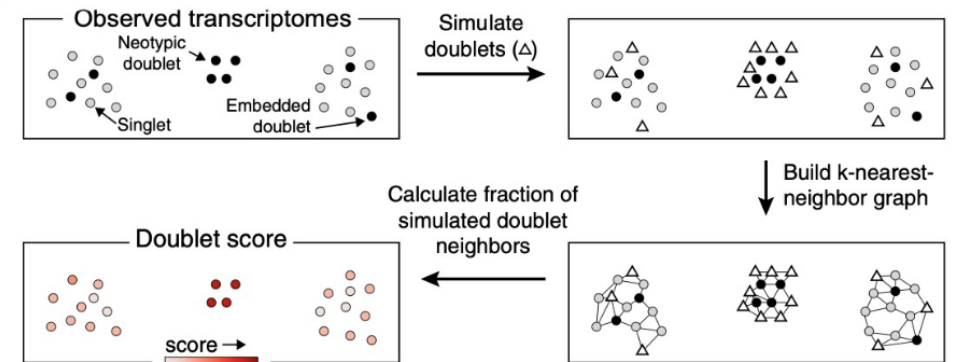
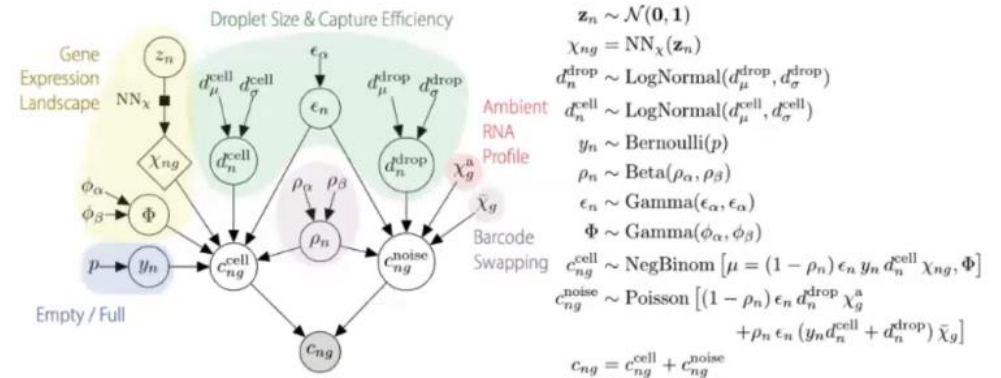
- Filtering cutoffs often differ between samples
- Visualise the distribution of the statistic to select a cutoff for filtering
- Alternatively filter cells empirically outside a specific range (3 median absolute deviations is common)
- The metrics are often correlated so bivariate scatterplots are useful to filter combinatorially



Ambient RNA correction and doublet detection

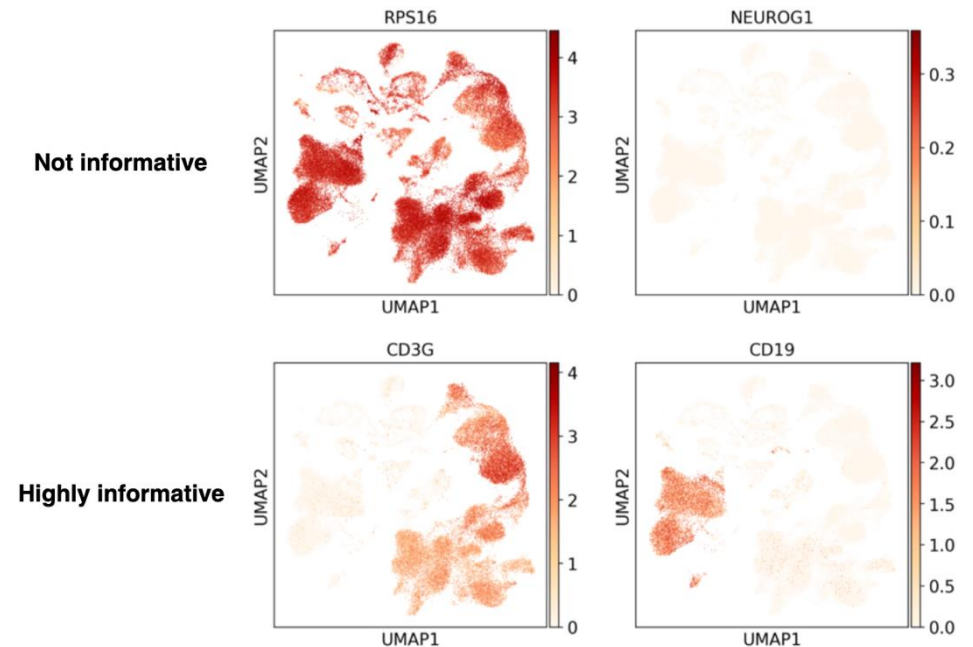
- Ambient RNA is **extracellular RNA in the suspension captured inside droplets** that contain other cells which can result in overestimation of expression of genes
 - Tools like SoupX and CellBender estimate the the ambient RNA profile and cell contamination fraction to subtract this from counts to get an adjusted value
 - They do not automatically improve data, and can overcorrect when ambient RNA contamination is low so should be used with caution
- Doublet detection tools **simulate artificial doublets** by combining expression profiles from pairs of cells
 - Each cell is measured for similarity to the synthetic doublets
 - Expected **doublet rate is often estimated from the number of loaded cells** and the highest-scoring cells satisfying this proportion are removed

h CellBender deep generative model



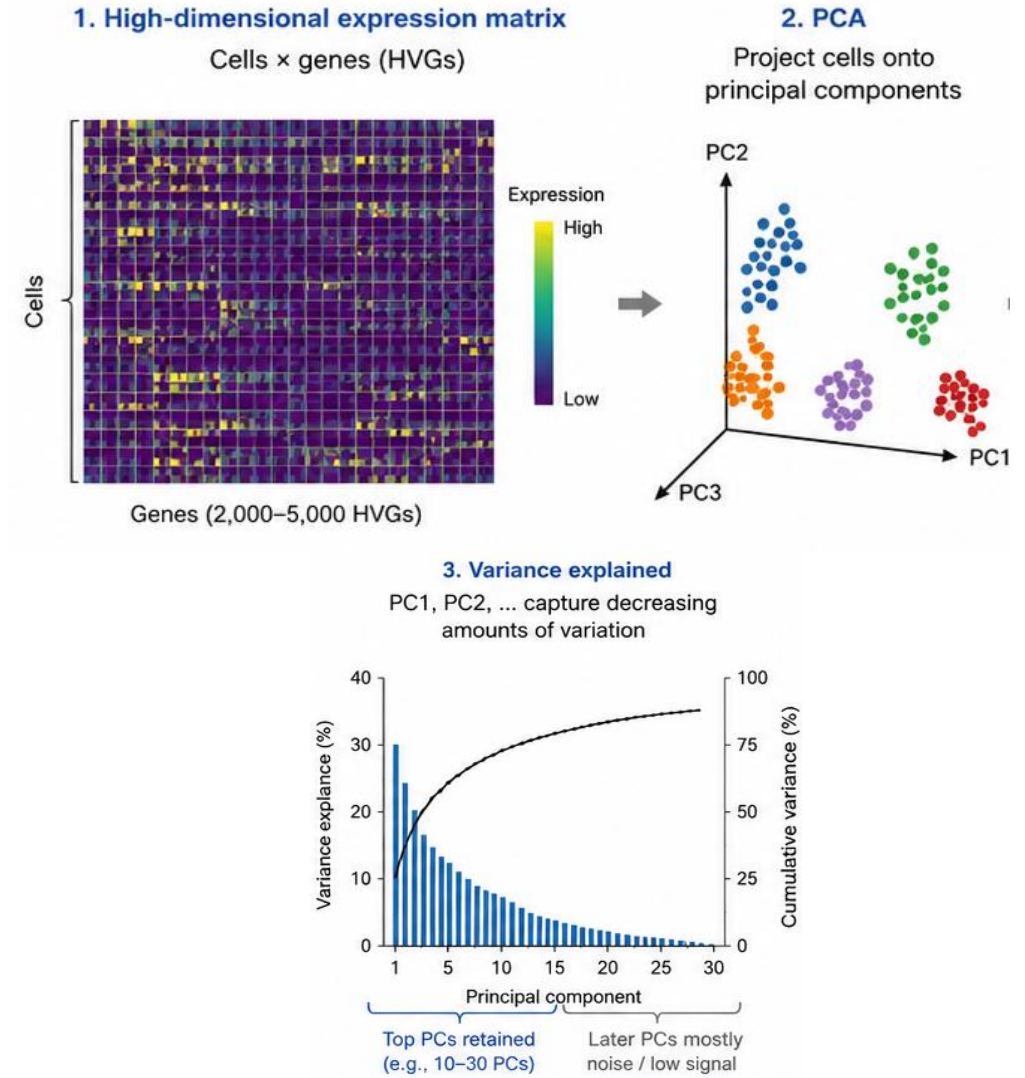
Normalisation and feature selection

- **Normalisation adjusts for differences in sequencing depth and RNA capture** to make expression values comparable across cells and samples
 - Most common approach is median count depth scaling with log plus one ($\log_1 p$) transformation.
- **Feature selection selects highly variable genes (HVGs)** which are genes that have greater variance than expected compared to similarly expressed genes
 - Normally 2000 - 5000 HVGs are selected level based on a variance-stabilising transform or Z-normalised dispersion value
 - By eliminating the weakly-informative genes which are not useful for capturing differences between cell types, states or conditions the signal-to-noise ratio is maximised
 - **NOTE: HVGs are not automatically biologically important; they are just useful for explaining variation in the dataset.**



Dimensionality reduction

- Even after reduction to 2000 – 5000 HVGs, **the count matrix will be still sparse** and subject to undesirable properties of high dimensional geometry.
- **Principal component analysis (PCA)** finds new axes that capture the greatest variance across cells.
 - The principal components represent orthogonal linear combinations of genes with the maximum variation
 - PCs do not usually correspond to one gene, but to a weighted expression programme involving many genes
 - There is a principal component for each dimension in the data, but the majority of the variation in data will be explained by a fraction of the original dimensionality.
 - In this way, low variation principal components can be discarded, effectively reducing the number of dimensions without sacrificing the variation in the data.
- The retained PCs are then used clustering, cell-type annotation and visualisation.

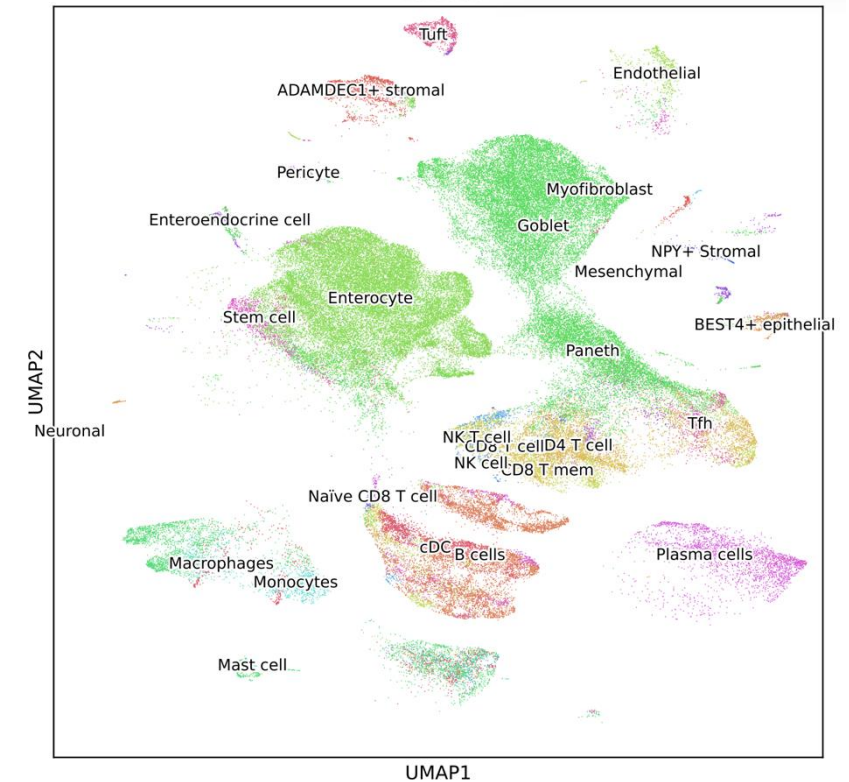


Clustering and cell type annotation

- **A K-nearest neighbour graph is constructed in the PC space**
 - Each cell becomes a node in the graph and edges connect cells that are close transcriptional neighbours.
 - Community detection algorithms such as Louvain or Leiden search for groups of cells that are more connected to each other than to the rest of the graph.
 - These communities are starting points for *de novo* annotation of cell types based on expression of canonical marker genes
 - Final labels often require several rounds of clustering and marker evaluation, and substantial domain knowledge
- **When a labelled reference dataset is available, a classifier can be trained**
 - Instead of manually labelling every cluster, a model learns expression patterns associated with known cell types.
 - Supervised classification trains a classifier on labelled cells, then predicts labels for new cells (CellTypist)
 - Deep generative models learn a latent representation while accounting for noise and batch effects and query data are projected into the learned latent space and labels are obtained by transfer learning (scANVI)

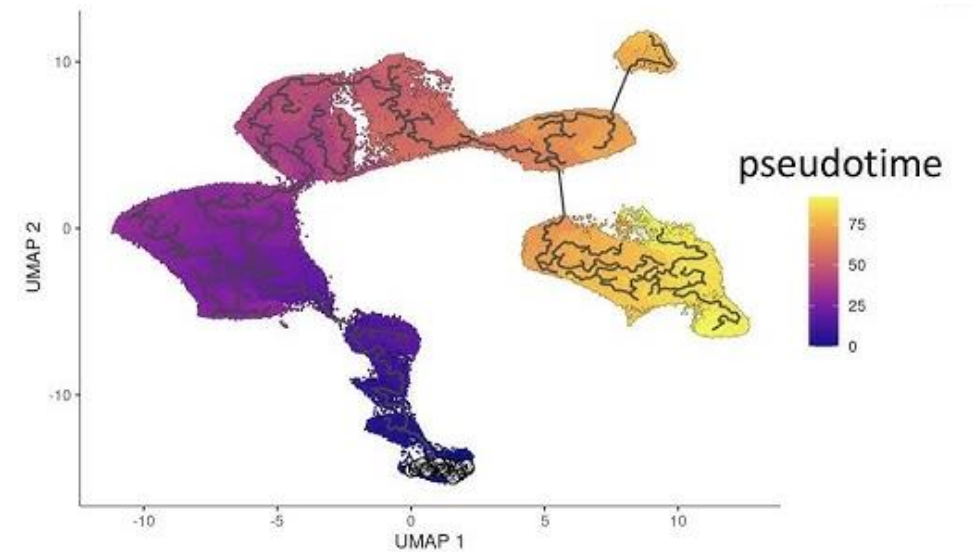
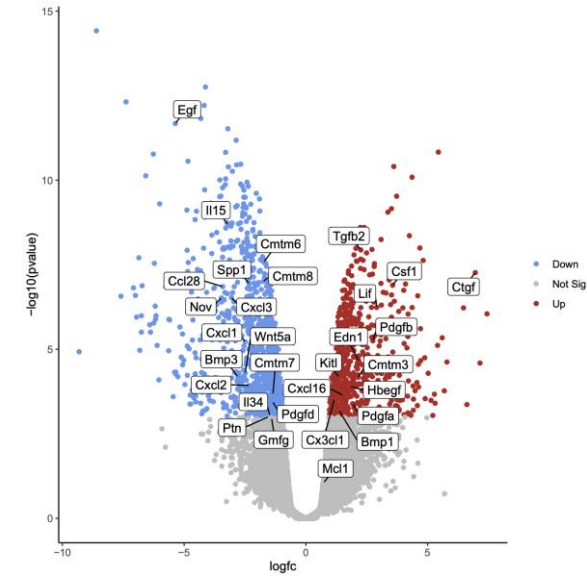
Visualising cells with UMAP

- **UMAP is a nonlinear, graph-based dimensionality reduction method** used to visualise single cell data
 - It constructs **a graph of similar cells in high-dimensional space**, then **optimises a low-dimensional layout** that preserves local neighbourhoods as well as possible.
 - **The size of gap between clusters is not directly interpretable** and the **axes have no biological units**
 - UMAP shape depends heavily on preprocessing choices, including HVG selection, number of PCs, batch correction, number of neighbours and minimum distance
- **UMAP embeddings are primarily a visualisation tool** to capturing non-linear dependencies in single-cell data and its embedding **are not to be used for downstream analysis**
 - **Nearby cells usually have similar transcriptomes** and distinct islands may represent different cell types,
 - Gradual transitions may suggest continuous biological processes.
 - Separation by batch or QC metrics may indicate technical artefacts.

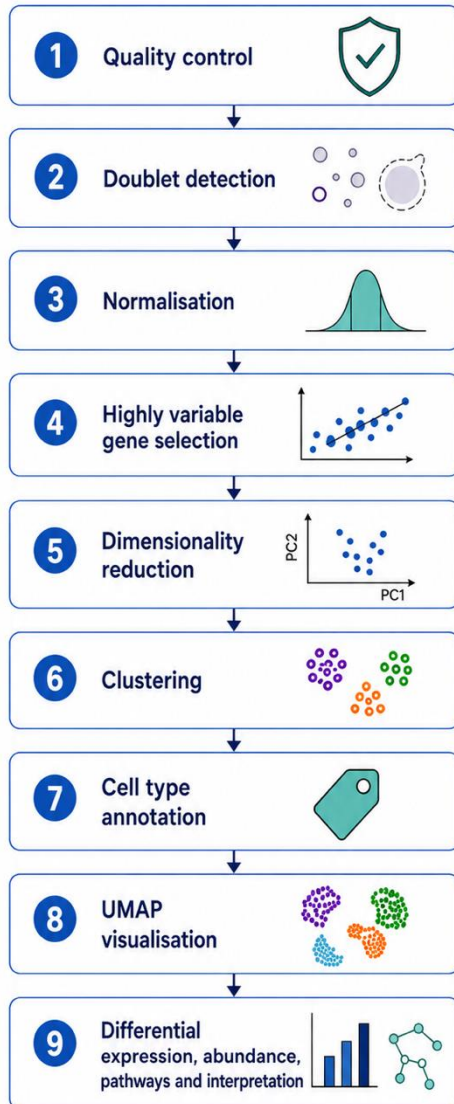


Downstream Analysis

- **Differential expression:** Identify genes that differ between cell types, samples or conditions
- **Differential abundance:** Quantify cell type proportions changes between samples or disease states.
- **Pathway and gene set analysis:** Summarise gene-level changes into biological processes.
- **Trajectory analysis:** Model gradual transitions such as differentiation, activation or regeneration.
- **Cell-cell communication:** Infer signalling interactions between cell populations.
- **Integration:** Combine scRNA-seq with other scRNA-seq datasets, or other modalities such as spatial transcriptomics, ATAC-seq, ChIP-seq proteomics or genotyping data



Roundup



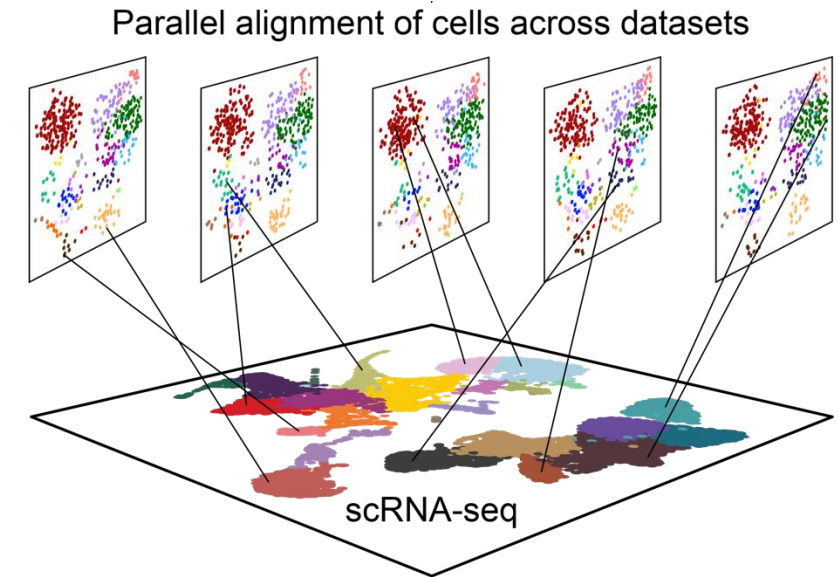
1. Filter low quality to remove technical artefacts
2. Predict and remove doublets
3. Scale gene expression values so cells are comparable
4. Select genes with informative variation across cells to focus downstream analysis on meaningful biological signal.
5. Densify matrix to capture the main patterns of variation between cells in a workable low-dimensional space
6. Group transcriptionally similar cells into units
7. assign biological identities to clusters based on marker gene expression or label transfer from a reference.
8. projects cells into two dimensions to help explore relationships between cell populations and identify broad structure in the data
9. Downstream analysis is used to determine how genes expression differ between conditions or sample groups across cell populations

Understanding integration and analysis of single-cell datasets

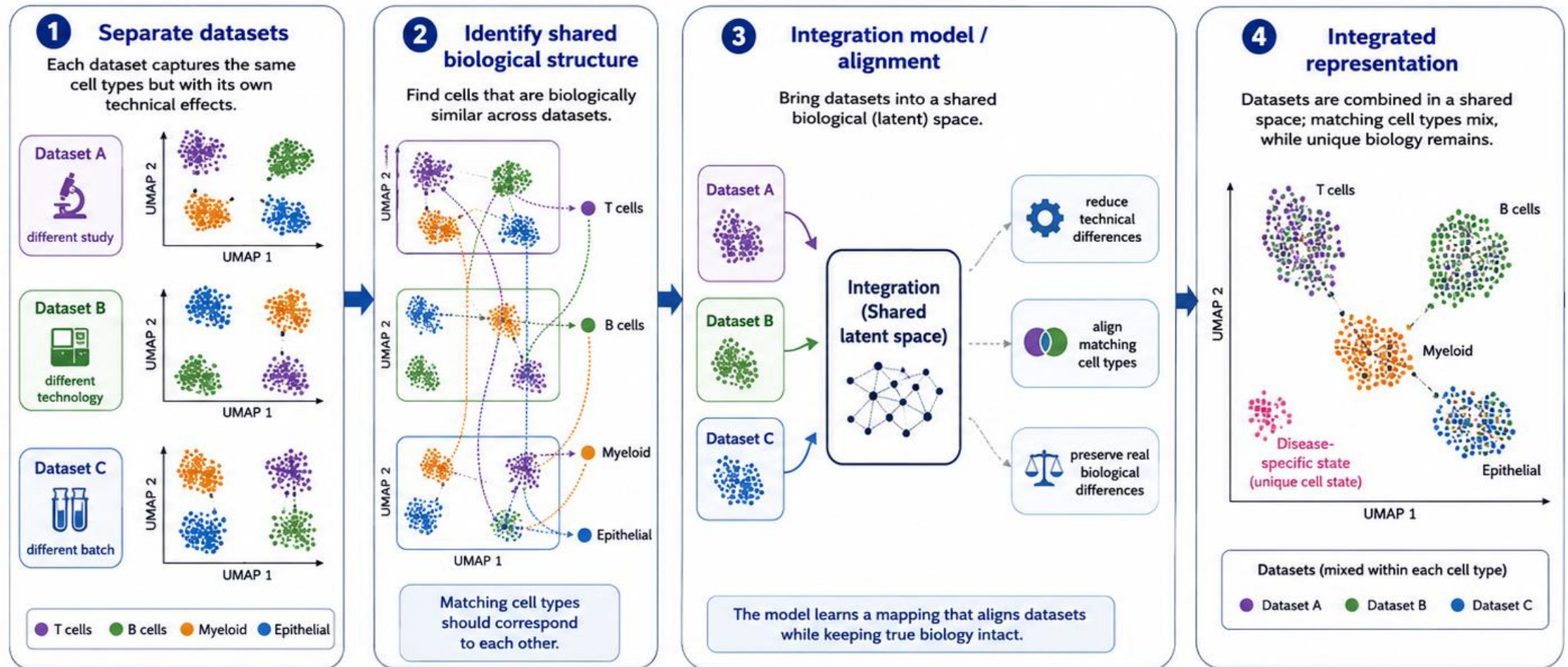


What is integration?

- Sometimes research questions require combined analysis of multiple dissimilar datasets
 - Pooling datasets across studies
 - Comparing biological behaviour across tissues, species or pathological states
 - Multi-omic analysis
- Merging cell embeddings from multiple datasets often results in cells clustering by dataset-specific variation making cross-sample comparison difficult
- Integration aims to combine multiple datasets with different into a shared space for joint analysis
- Whereas batch correction uses covariates to regress out unwanted technical variation within datasets, integration aims to align comparable biology while preserving true biological differences between datasets
 - The aim is **not always to remove all differences between datasets but to make cells comparable while preserving meaningful biological structure.**
 - Integration is therefore a balance between removing technical variation and protecting meaningful biology.

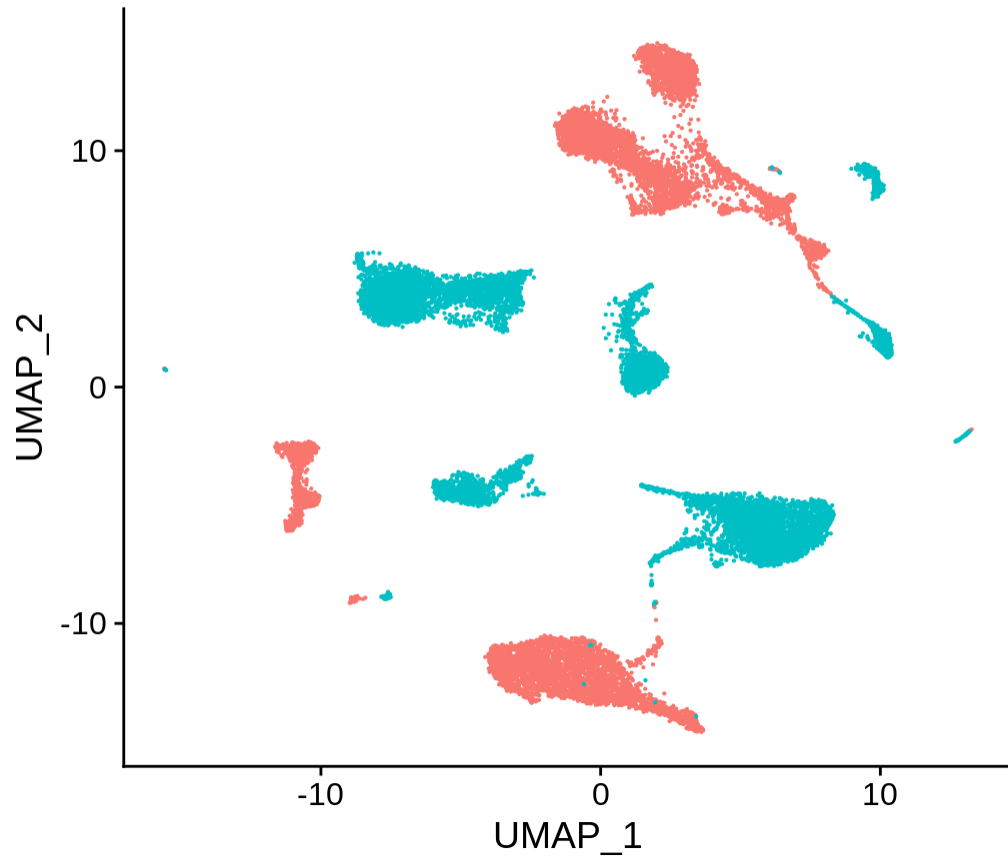


What integration does

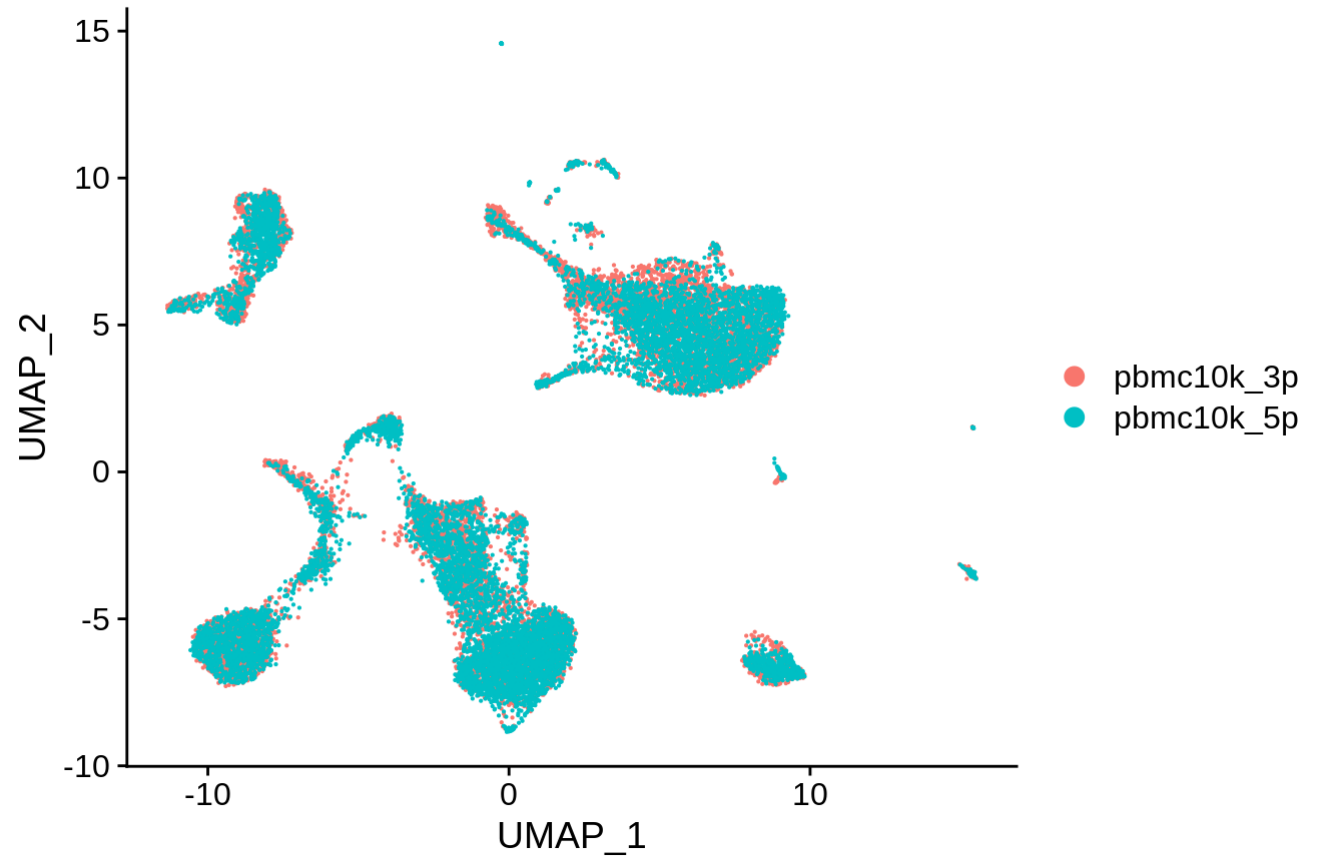


Integration - before and after

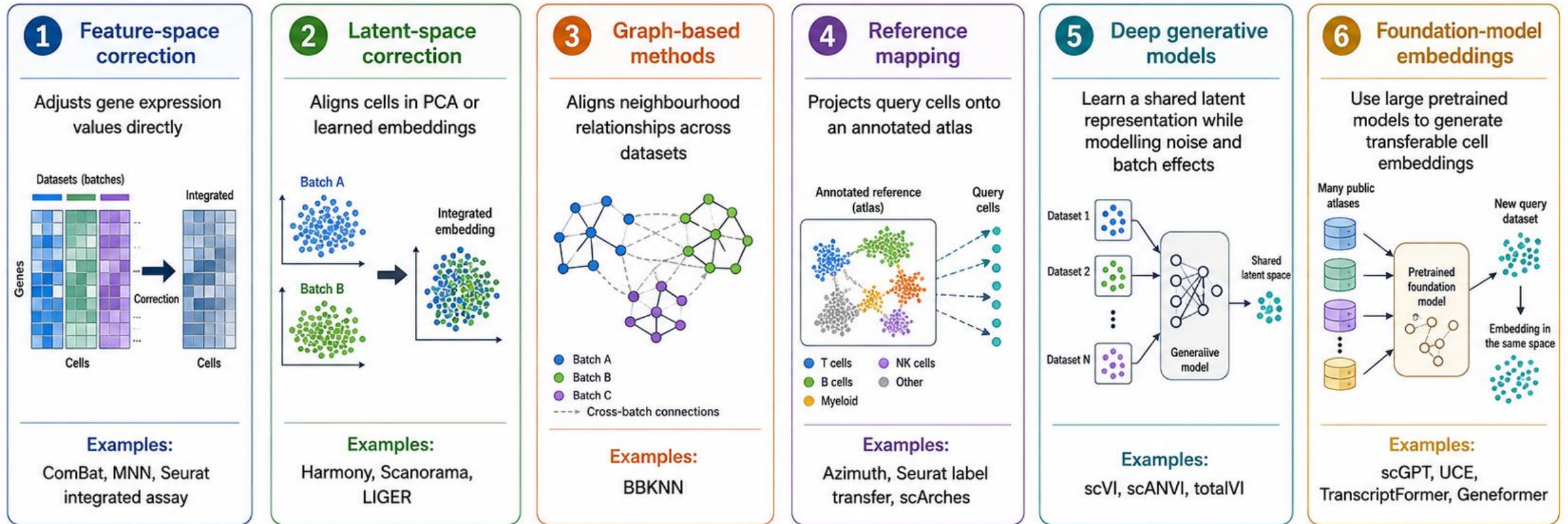
10k 3' PBMC and 10k 5' PBMC cells, before integration



10k 3' PBMC and 10k 5' PBMC cells, after integration (Seurat 3)



Integration strategies for single-cell transcriptomics

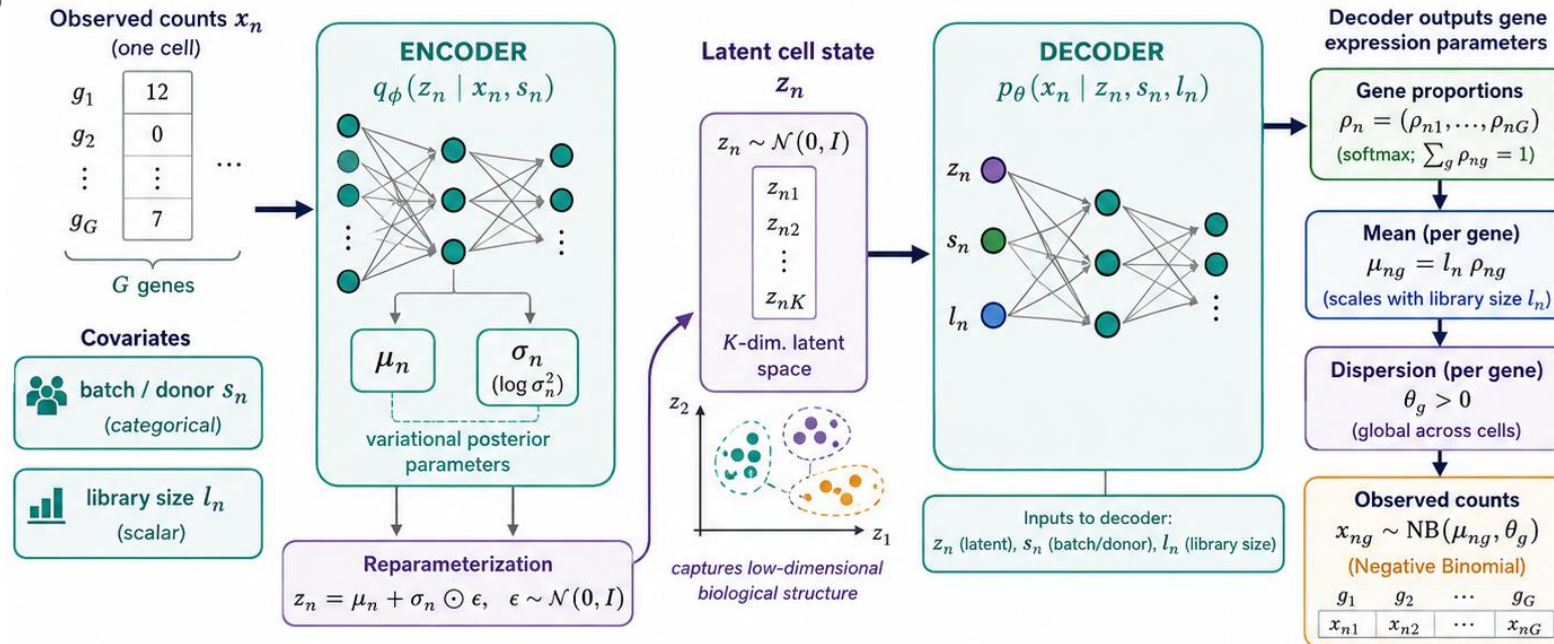


Latent space correction (Harmony, Scanorama, LIGER)

- **PCA embeddings are adjusted to reduce dataset-specific effects** so that biologically equivalent cells from different datasets are positioned closer together.
 - **Harmony** does this iteratively through rounds of soft clustering and embedding correction until convergence
 - **Scanorama** identifies shared cell types between pairs of datasets and estimates how corresponding cells should be aligned by a mutual-nearest-neighbour approach
 - **LIGER** uses integrative NMF to decompose datasets into shared and dataset-specific factors, aiming to identify components of cell identity that are common across datasets
- **Raw or normalised gene expression values are usually left unchanged** preserving the original gene-level signal for marker gene analysis and differential expression testing.
- **Best suited for pooling datasets with broadly matching cell types** where batch effects dominate
- **Susceptible to overcorrecting disease states**, rare populations or condition-specific biology if these are confounded within batch effect structure.

Single-cell Variational Inference

- Variational autoencoders (VAEs) are probabilistic neural networks that learn a low-dimensional latent representations of data where unwanted variation are regressed out as covariates in the model
- scVI is a VAE-based model designed specifically for single-cell RNA-seq count data
 - **Encoder**, which compresses each cell's gene expression profile into a low-dimensional latent representation
 - **Decoder**, which tries to reconstruct the original gene expression counts from that latent representation

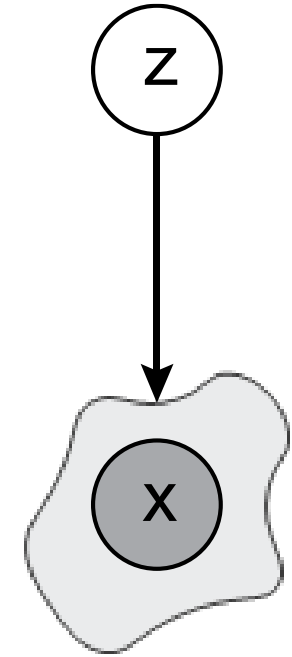


scVI training

- VAEs **scale very well to large datasets** and **provide uncertainty estimates** for each cell which is encoded as a probability distribution in latent space rather than a single fixed point.
- They **learn nonlinear latent representations** that can capture complex cell states and handles sparsity, overdispersion and count noise probabilistically.
- Requires training so results depend on model setup, covariate choice, training parameters and data quality.
- Optimising and validating models is time consuming and requires careful tuning of model architecture and training settings
- Parameters depend on the dataset size, complexity and integration goal:
 - n_latent: dimensionality of the learned latent space
 - n_layers: depth of encoder/decoder networks
 - n_hidden: number of neurons in hidden layers
 - dropout_rate: regularisation strength
 - gene_likelihood: count model, usually negative binomial or ZINB
 - max_epochs: number of training rounds
 - batch_size: number of training samples propagated
 - learning_rate: how fast the neuron weights change between epochs
- Evaluate parameters by checking:
 - validation loss / ELBO
 - preservation of rare or disease-specific states
 - batch mixing
 - cell type separation
 - marker gene consistency

The scVI ecosystem

- **scvi-tools** has a large ecosystem of specialised VAEs
- **scANVI** Semi-supervised extension of scVI.
 - Uses known labels from reference cells to annotate unlabelled query cells.
 - Useful for label transfer and reference mapping.
- **SysVI** Designed for cross-system or cross-platform integration.
 - Useful where technology or experimental system effects are strong.
- **TotalVI** Jointly models RNA and protein abundance.
 - Useful for CITE-seq or single-cell multiomic data with antibody-derived tags.
- **MultiVI** Integrates scRNA-seq and scATAC-seq.
 - Useful when some cells have RNA, some have ATAC, or some have both.
- **scArches** Enables query-to-reference mapping by transfer learning.
 - Useful for projecting new datasets into a pretrained reference model.



Foundation model embeddings (scGTP, UCE, TranscriptFormer)

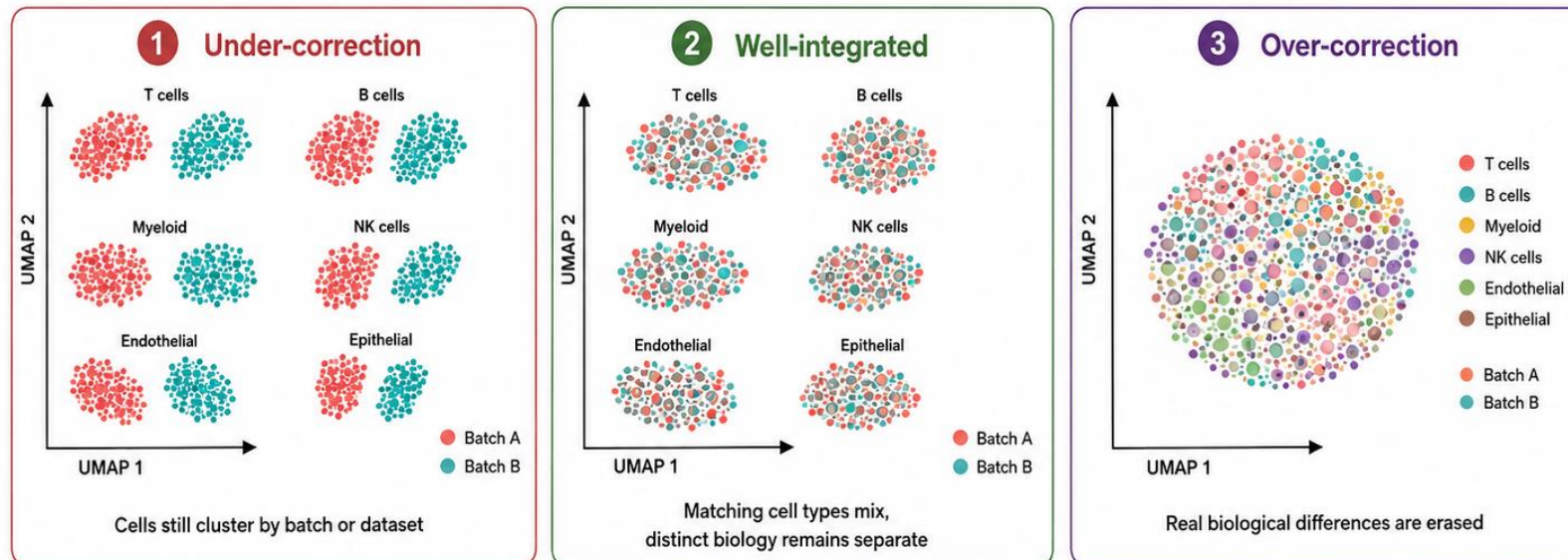
- Single-cell foundation models are large pretrained models designed to learn general representations of cells and genes from large-scale single-cell datasets.
 - These models are trained on millions of cells from many tissues, organs, developmental stages, disease contexts and species
 - They are inspired by transformer models from natural language processing, where tokens are learned in context, with genes treated as tokens and cells as gene-expression profiles as “sentences”
- Instead of training a model from scratch for each new dataset, foundation models are pretrained on enormous reference atlases to learn generalised patterns such as
 - which genes tend to be co-expressed
 - which gene programmes define cell states
 - how cell types relate across tissues
 - how perturbation, activation or differentiation states may appear transcriptionally
- Once pretrained, the model can be reused to generate general-purpose cell or gene embeddings for new datasets.
- This can be useful when labelled data are limited, because the model brings in information learned from large external reference datasets.

Using single-cell foundation models

- **Zero-shot use applies the pretrained model directly** to a new dataset
 - The model has already learned general patterns from large reference atlases, so it can generate embeddings without retraining on the query dataset.
 - Zero-shot use may not represent a specific tissue, platform or disease state well if it is rare in the training data.
- **Fine-tuning takes the pretrained model and updates retraining it slightly** to better represent a specific query dataset.
 - Useful when the query data differs substantially from the pretraining data.
 - Requires careful validation, because the model may overfit to the query dataset or learn dataset-specific artefacts.
- Best viewed as transfer-learning tools **which leverage enormous reference atlases to generalise understanding of new data**
- This still an emerging area, so performance can vary and it is not yet safe to assume that a model trained on large atlases will generalise equally well to every biological condition

Integration Evaluation and Common Failure Modes

- **Batch mixing:** known cell types should contain cells from multiple datasets or batches.
 - Compute iLISI, kBET, batch ASW, graph connectivity, PCR batch score
- **Cell identity and biological separability:** distinct cell types and biological states of interest should remain separated after integration.
 - cell type ASW, cLISI, NMI, ARI, isolated label score
- **Sensible markers:** cell types should still express expected marker genes before and after integration.
- **Rare populations retained:** rare but real cell types should not be merged into more common groups.



Final Points

- Integration is used when the **research question requires comparing or combining multiple datasets**.
- The goal is not to make all cells overlap, but to **align biologically equivalent cells across datasets** while meaningful differences remain interpretable.
- **Good integration removes unwanted batch effects while preserving true biological variation.**
 - Overcorrection can erase disease states, rare populations or condition-specific biology.
 - Undercorrection leaves cells clustering by batch, platform or study.
 - Integration must be evaluated using both technical metrics and biological validation.
- Gene-level testing should still use original counts or normalised expression, not corrected embeddings.
- Choose an integration strategy based on need; **more sophisticated approaches require more effort** to execute well, and **simpler techniques are not inherently worse**.